



Software Inspections - 2



Lecture # 16

Recap

- Introduced software inspections as very important defect removal technique
- Made a very strong case with the help of published data that software inspections reduce the overall cost and schedule of the software project
- They have many ancillary benefits as well: employee morale, customer satisfaction, shorter time to market, as well as long-term financial strength of the organization. We'll discuss them today also

Today's Lecture

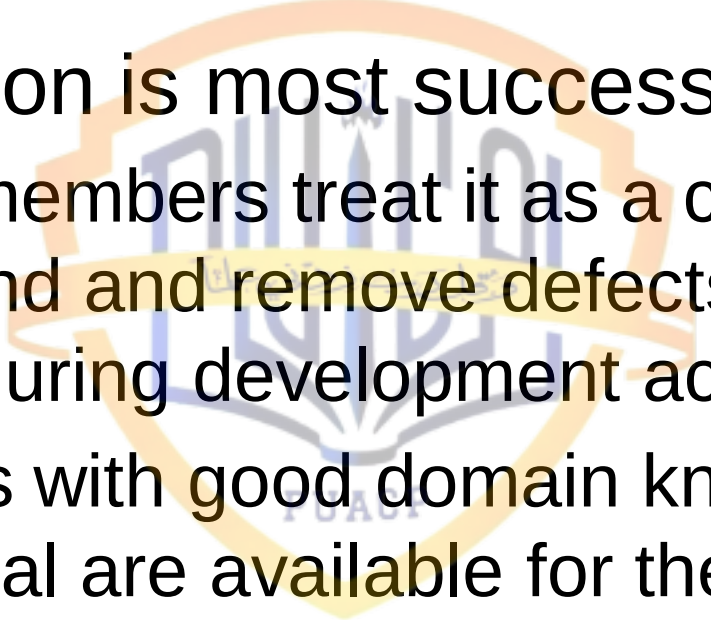
- We'll continue our discussion on software inspections
- We ended our last lecture on the pre-conditions for software inspections, and we'll today start our discussion with some success factors that worked for many companies

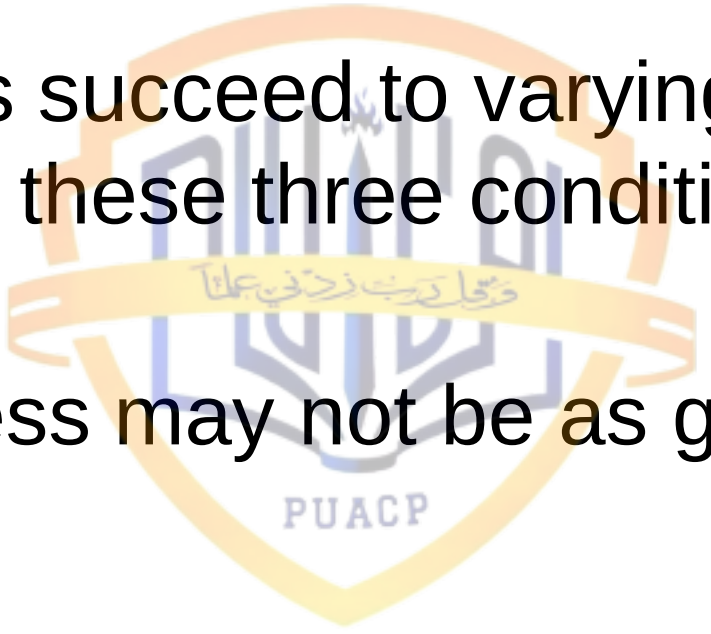
Success Factors - 1

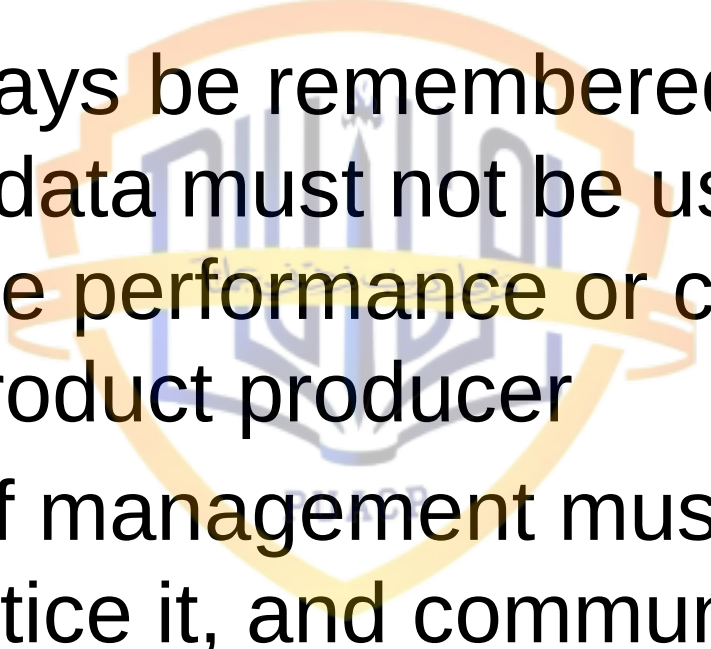
- Kept to the basics
- Trained teams rather than individuals
- Established a policy that inspections are safe
- Followed the proven method, before adapting or tailoring it
- Gave proper time for inspections to take root

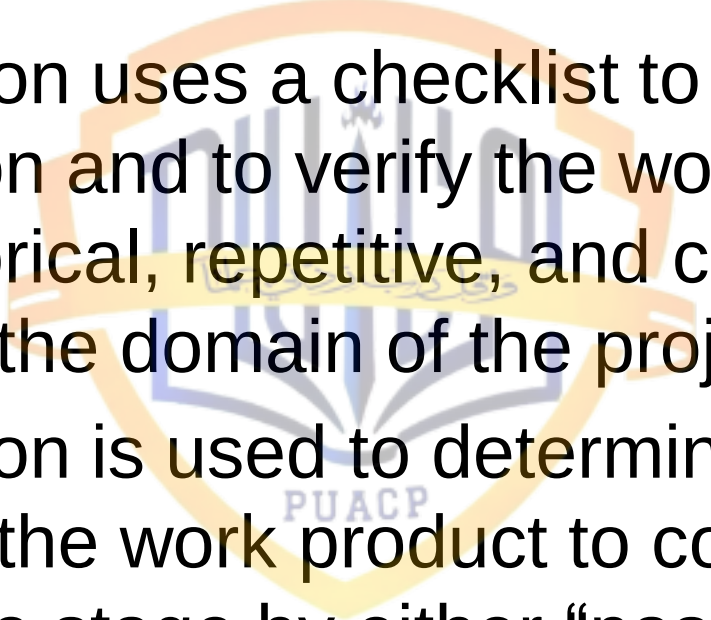
Success Factors - 2

- Analyzed and used the data resulting from inspections
- Built on their own successes
- Learned what was not necessary to inspect
- Rewarded the performance of inspections
- Shared the success
- Allocated budget and time for inspections

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- An inspection is most successful when
 - All team members treat it as a cooperative effort to find and remove defects as early as possible during development activities
 - Inspectors with good domain knowledge of the material are available for the inspection
 - Inspectors are trained in the inspection process

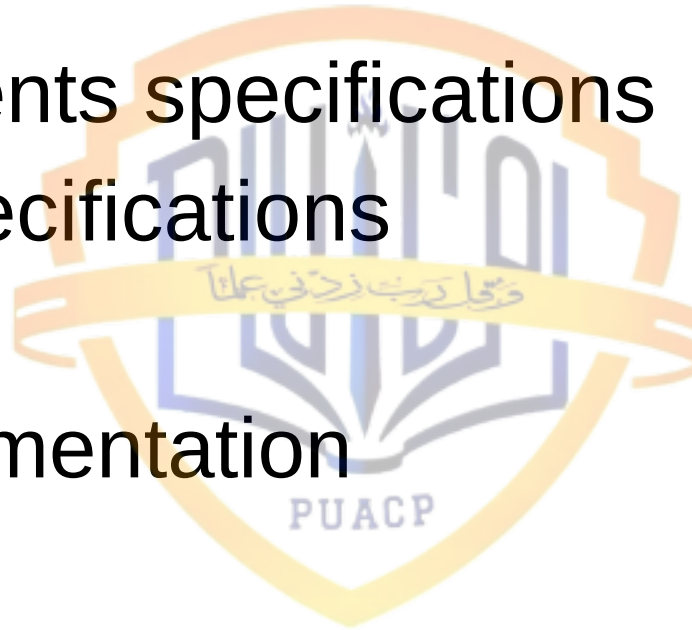
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- Inspections succeed to varying degrees even when these three conditions are not met
 - Effectiveness may not be as good

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- It must always be remembered that inspection data must not be used to evaluate the performance or capability of the work product producer
 - All levels of management must accept this value, practice it, and communicate the commitment to all personnel

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- The inspection uses a checklist to prepare for the inspection and to verify the work product against historical, repetitive, and critical defect types within the domain of the project
 - The inspection is used to determine the suitability of the work product to continue into the next life cycle stage by either “passing” or “failing”
 - If the work product fails, it should be repaired and potentially re-inspected

Work Products

- Requirements specifications
- Design specifications
- Code
- User documentation
- Plans
- Test cases
- All other documents



Inspection Steps - 1

■ Overview

- Provides the inspection participants a background and understanding, when warranted, of the scheduled inspection material

■ Preparation

- Allows time for the inspection participants to sufficiently prepare for the inspection meeting and list potential defects

Inspection Steps - 2

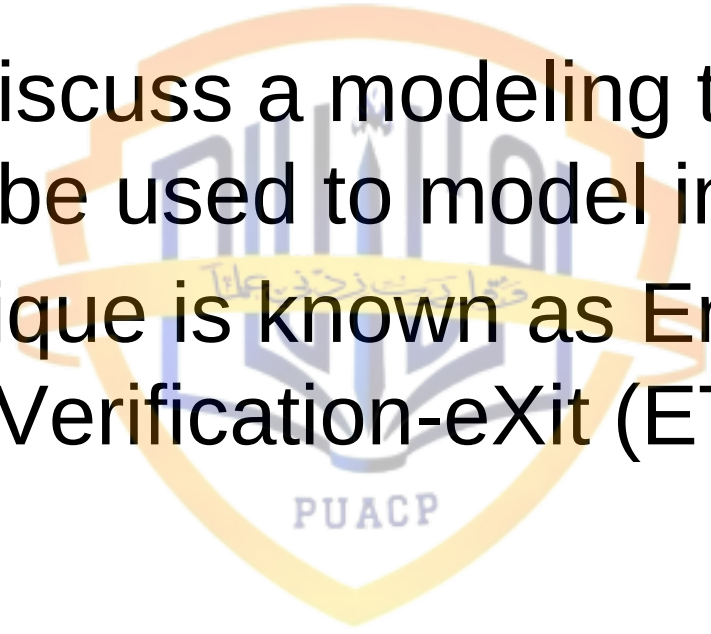
- Inspection meeting
 - Identifies defects before work product is passed into the next project stage
- Rework
 - Fixes identified defects and resolves any open issues noted during the inspection
- Follow-up
 - Verifies that all defects and open issues have been adequately fixed, resolved, and closed out

Other Inspection Steps - 1

- Planning and scheduling
 - To ensure adequate time and resources are allocated for inspections and to establish schedules in the project for work products to be inspected, to designate the inspection team, and to ensure the entry criteria are satisfied
- Data recording
 - To record the data about the defects and conduct of the inspection

Other Inspection Steps - 2

- Analysis meeting
 - Which is held after the inspection meeting, to begin defect prevention activities
- Prevention meeting
 - Which is held periodically after sets of inspections have been performed to determine probable causes for selected defect types, instances, or patterns

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- let's now discuss a modeling technique, which can be used to model inspections
 - This technique is known as Entry-Task-Validation/Verification-eXit (ETVX) technique

ETVX Representation

- The model expressed as a set of interconnected activities each of which has four sets of attributes
 - Entry (E)
 - Task (T)
 - Validation/Verification (V)
 - Exit (X)

Entry

- The Entry section defines the entry criteria that must be satisfied for the process to be initiated, and list the work products that must be available as inputs to the process

Tasks

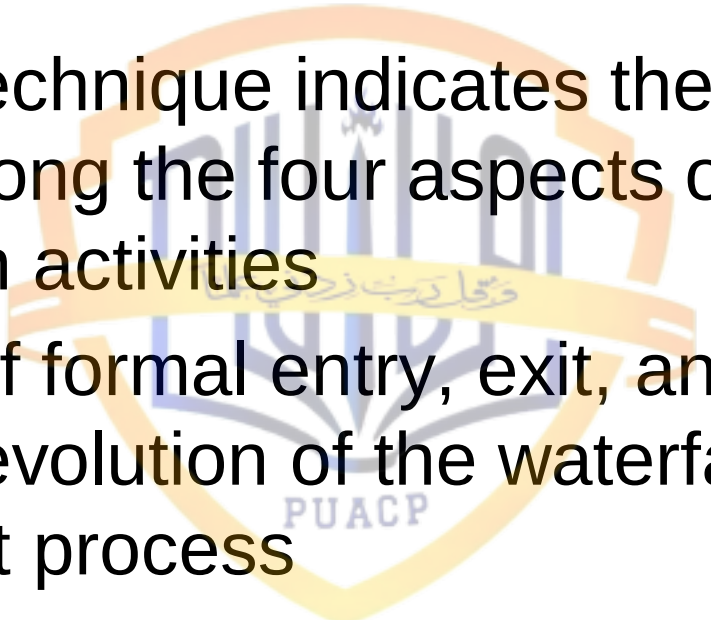
- The Task section defines work to be carried in performing the process. The order of the task is generally, but not strictly sequential. Some tasks may be concurrent with other tasks

Validation/Verification

- The validation/verification section defines steps for validating/verifying that the process has been properly executed, and that the associated work products meet project objectives

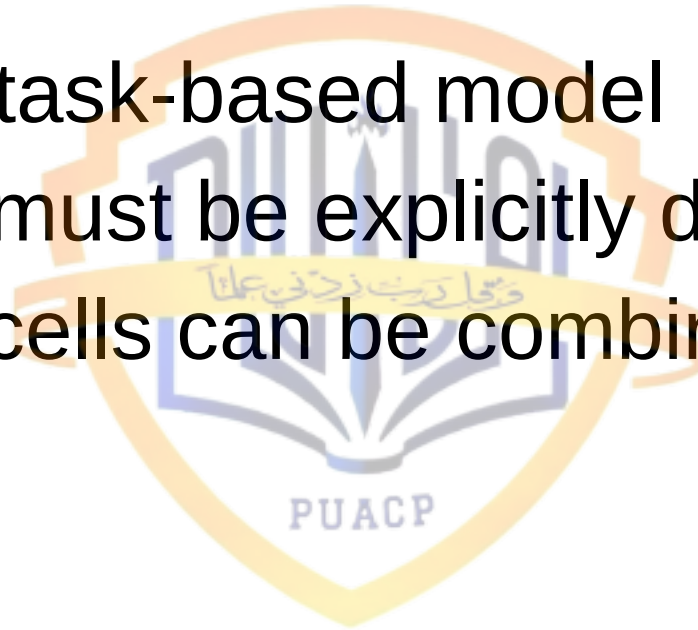
Exit

- The Exit section defines the exit criteria that must be satisfied for the process to be terminated. The exit criteria usually define completion and verification work products, in terms of qualitative aspects of the products

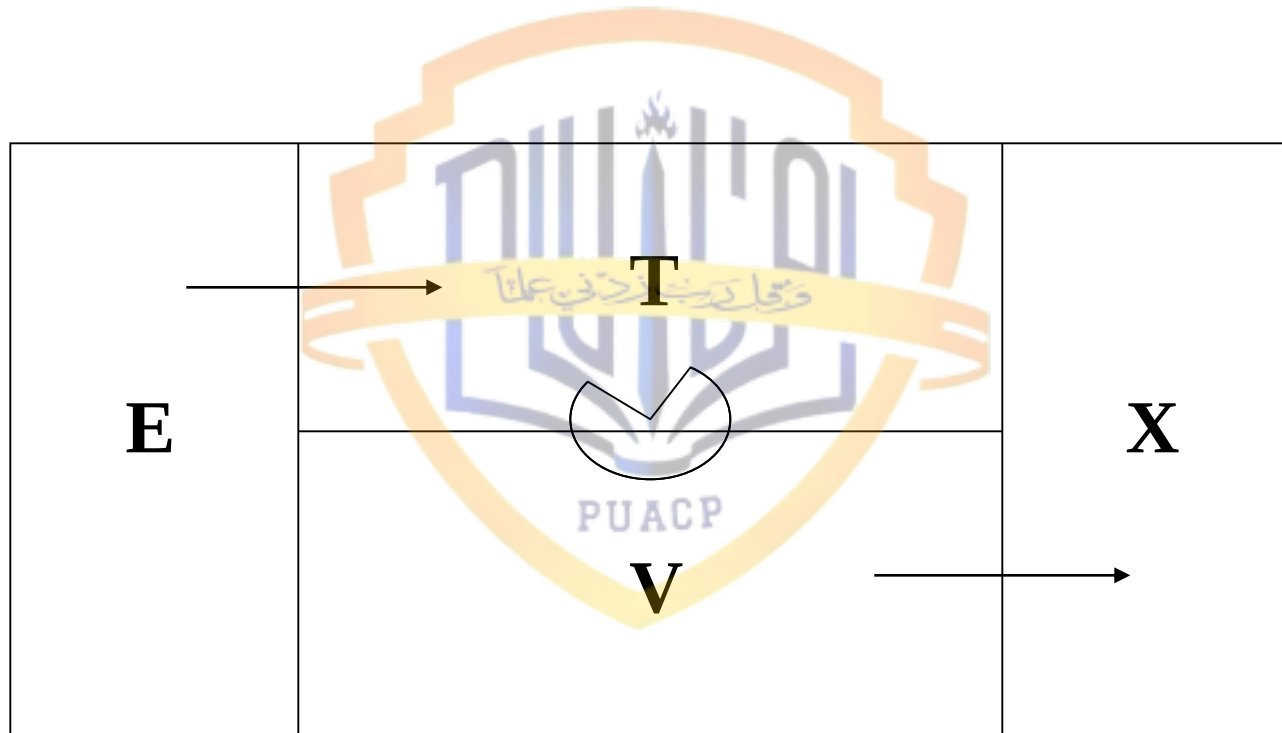
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- The ETVX technique indicates the relationship and flow among the four aspects of an activity and between activities
 - The notion of formal entry, exit, and criteria go back to the evolution of the waterfall development process
 - The idea is that every process step, **inspection**, function test, or software design has a precise entry and exit criteria

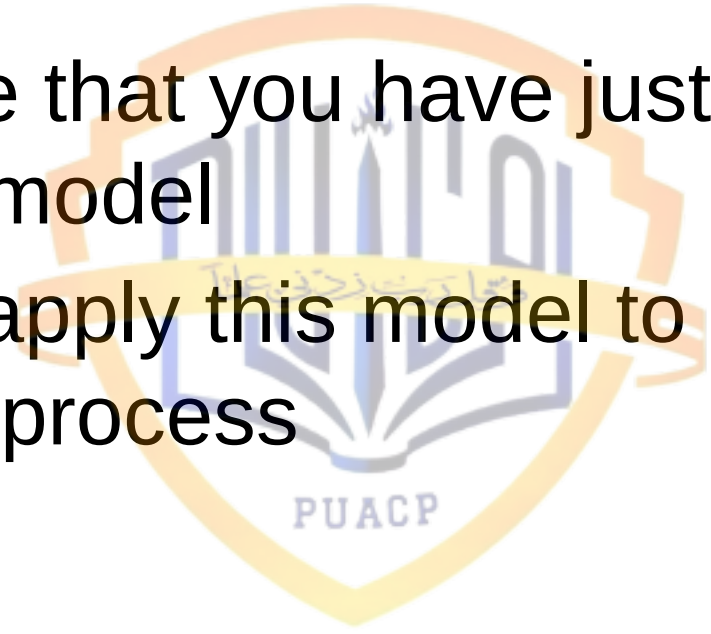
Characteristics of ETVX

- ETVX is a task-based model
- Each task must be explicitly defined
- The basic cells can be combined to create process

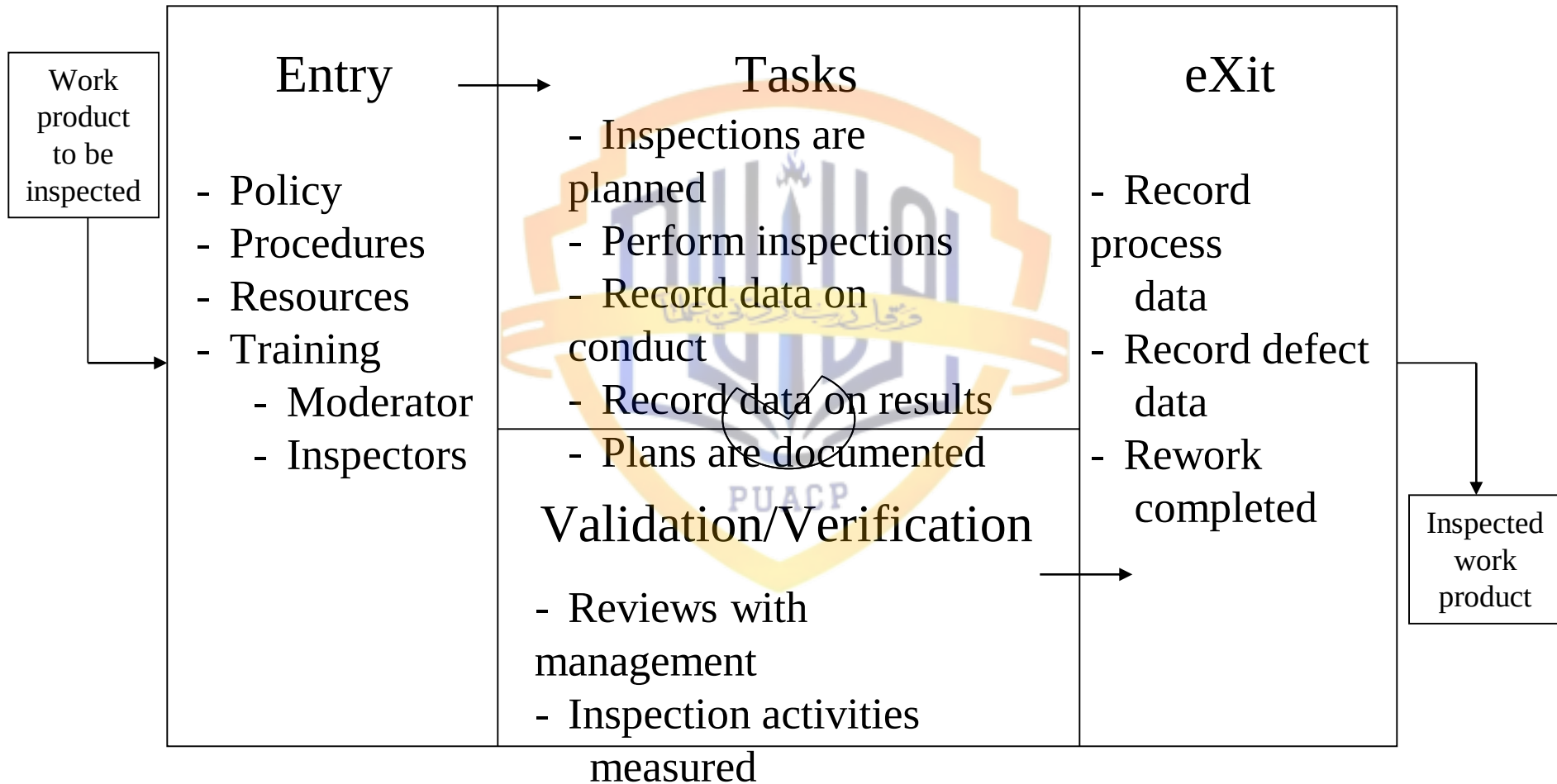


The ETVX Process Definition Paradigm



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- The picture that you have just seen shows the ETVX model
 - Let's now apply this model to the inspection process

Practices in the Inspection Process



Ancillary Purposes

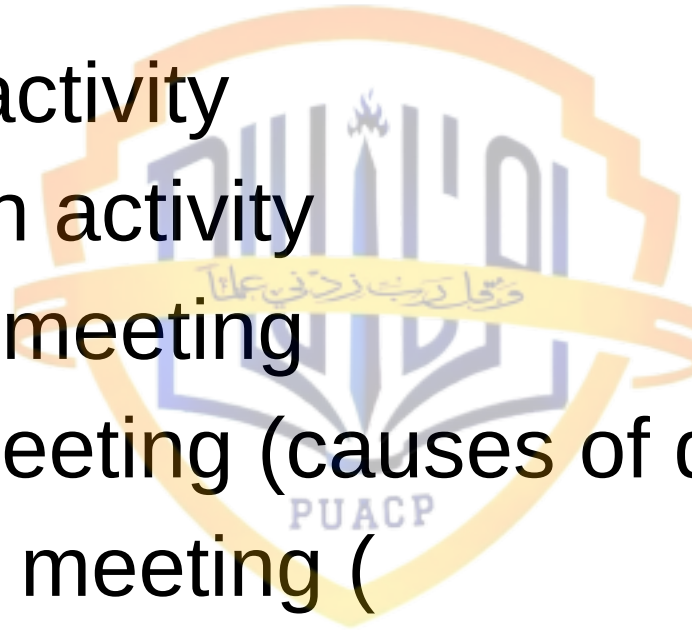
- Improvement in productivity
- Education and increased knowledge sharing
- Developing backup/replacement capability
- Process improvement
- Early product quality visibility
- Product re-development
- Building team spirit

Productivity Improvement

- Fagan calculated a 23% productivity improvement during the VTAM study – one of the first projects in which inspections were used
- “An improvement in productivity is the most immediate effect of purging errors from the product”
 - Michael Fagan
- “Inspection reduces the development cost during test by 50%” based on IBM studies
 - Norris

Education and Increased Knowledge Sharing

- Overview activity
- Preparation activity
- Inspection meeting
- Analysis meeting (causes of defect)
- Prevention meeting (



Back-Up/Replacement Capability - 1

- Many organizations have high turnover rates, and in many cases only a few people (or even one person) has the required knowledge of a product or key parts of a product
- Where turnover is high, knowledge can literally walking out of the door
- To mitigate this risk, some organizations have elected to inspect 100% of all work products

Back-Up/Replacement Capability - 2

- Basically, they are providing backup, and this is 'dynamic backup'
- In these situation, inspections are used to spread the knowledge as fast and as far as possible
- This education also provides a flexibility to react quicker when there are customer needs

Back-Up/Replacement Capability - 3

- In some situations, maintenance of the work product may be transferred to a new organization or a subcontractor
- So new people need to be educated and trained on the work products as fast as possible
- So, inspections are used to create backups or replacement owners of work products

Back-Up/Replacement Capability - 4

- The choice for when to consider these types of inspections is determined by
 - Defect backlogs
 - Change request backlogs
 - Possibilities for re-engineering
 - Risk mitigation for volatile product sections
 - Turnover rates
 - Recruitment rates

Back-Up/Replacement Capability - 5

- To be successful for in these inspections, the author of the work product has to be present
- “Inspections broaden the knowledge base of the project in the group, create potential backup programmers for each module, and better inform the testers of the functions they are to test”
 - Norris

Process Improvement

- Data is gathered during the inspection and later analyzed to understand the process of doing the inspection and later to improve it
- (Say something on time availability)

Early Product Quality Visibility

- Quality of the work product and that of the software product starts to become clear in the early stages of the software development life cycle

Product Re-Development

- Products with multiple releases can have high volumes of changes in some areas
- And, some work products with high defect rates may have to be re-engineered
- Inspections are “a very good mechanism for highlighting and prioritizing candidate areas for enhancement”

Building Team Spirit

- “The review process also promotes team building. It becomes one of the first steps toward establishing a good development team, by substituting an environment where programmers work alone throughout their careers, for a programming team environment in which each individual feels free to discuss and critique everyone else’s program. Implicit in the concept of a team is the notion of working closely together, reading each other’s work, sharing responsibilities, learning each other’s idiosyncrasies both on technical and personal level, and accepting altogether as a group shared responsibility for the product where each member can expect similar rewards if the project is a success and similar penalties if the project fails”

Where Do Inspections Fit in Software Development Process?



The Cost of Inspections - 1

- There is a cost for every activity during project life cycle. The cost is determined by many factors:
 - Capability of programmers performing the activity
 - The defined process for the activity
 - Stability of the input
- Inspections have a cost also

The Cost of Inspections - 2

- Time spent on inspections is an investment. You inspect now, you invest now, and then you reap the benefits down the line
- Concern should only be when the inspections are performed for the first time
- Once the cost question is removed from management's thinking, the time needed up front in a project is no longer a concern

Cost in Hours/Defect During Code Inspections

COMPANY	HOURS/DEFECT
AT&T	.67 – 1.4
ICL	1.2 – 1.6
JPL	1.5 – 2.1
Bell-Northern	Less than .4
HP	.2
Bull	1.43
ITT Industries	~1

References

- Software Inspections by Ronald A. Radice, Tata-McGraw Hill, 2003 (Chapter 2)
- Software Engineering: A Practitioner's Approach by Roger S. Pressman